

Mean, Median, and Mode

Lesson 8-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*Mean of 2, 4, 6: add them ($2+4+6=12$), divide by 3
→ mean = 4*

Mean

The average. Add all the numbers, then divide by how many there are.

*Data: 1, 3, 5, 7, 9 → the middle (3rd) value is 5, so
median = 5*

Median

The middle number when you put them in order.

*Data: 2, 3, 3, 5, 7 → 3 appears twice (most often),
so mode = 3*

Mode

The number that shows up most often.

*Data: 10, 12, 11, 13, 50 → 50 is far from the others,
so 50 is an outlier*

Outlier

A number that is much bigger or smaller than the rest.

*If most values cluster in the center with fewer at the
ends, the distribution is symmetric*

Data distribution

How the data looks: where it sits and how spread out it is.

*8, 9, 10, 11 has low variability (close together); 2,
10, 25, 40 has high variability (far apart)*

Variability

How spread out the numbers are.

Key Ideas & Notes

- You're the stats analyst for the school basketball team.
- Coach needs a report on the team's scoring: in the last 7 games they scored 42, 38, 55, 41, 38, 44, and 40 points.
- Which average best represents the team's typical performance?

Think About It

- What are the highest and lowest scores?
- Do any scores repeat?
- Which score seems like a typical game for this team?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The team scored 42, 38, 55, 41, 38, 44, 40 points over 7 games. Which 'average' would best tell the coach the team's typical scoring: mean, median, or mode?

Level 1 support

Start here: Mean, median, and mode each describe the center of the data in a different way.

I would use the ____ because it best shows the typical score.

Usaría la ____ porque muestra mejor el puntaje típico.

The ____ is the most common score, which is ____.

La ____ es el puntaje más común, que es ____.

Word bank: mean median mode average data set

Listen for: Students distinguish the three measures and justify a choice, noting the mode (38) is the most repeated and the mean/median sit near the center.

Level 2

Justify: which measure would change the most if one game's 55 were actually 100?

Explain why.

- The ____ would change most because it uses ____.
- The mode and median would stay the same because ____.

EXPLORE

After ordering the scores on the number line, how do you find the median, and how is it different from the mean you calculated?

Level 1 support

Start here: The median is the middle of ordered data; the mean is the equal-share total.

To find the median, I first ____ the scores, then pick the ____.

Para hallar la mediana, primero ____ los puntajes y luego elijo el ____.

The mean is different because I have to ____ and ____.

La media es diferente porque tengo que ____ y ____.

Word bank: mean median mode order data set

Listen for: A strong answer orders the data, finds the middle value as the median, and explains the mean is found by summing all values and dividing.

Level 2

Compute and compare: find the mean and median of 42, 38, 55, 41, 38, 44, 40. Are they close? What does that tell you?

- The mean is ____ and the median is ____.
- They are close/far apart, which tells me the data is ____.

CONNECT

Quiz scores were 85, 92, 78, 88, 92, 95, 72, 88, 92, 84. The teacher says 'most scored around 87'; a student says 'the most common score was 92.' Who used the mean, and who used the mode?

Level 1 support

Start here: Different people pick different measures of center to make their point.

The teacher used the _____, and the student used the _____.

El maestro usó la _____, y el estudiante usó la _____.

The _____ is more useful here because _____.

La _____ es más útil aquí porque _____.

Word bank: mean median mode average data set

Listen for: Students identify 87 as the mean and 92 as the mode, and argue which is more useful for describing the class's typical performance.

Level 2

Critique: is the student technically right that 92 was 'the most common,' yet possibly misleading about how the class did? Defend.

- The student is right that the mode is _____, but it can mislead because _____.
- A fairer summary of the whole class is the _____ since _____.

REFLECT

For the data set 4, 7, 10, 7, 12, talk through how you find the median.

Level 1 support

Start here: Order the values, then find the middle one.

First I order the values: ____.

Primero ordeno los valores: ____.

The median is ____ **because it is in the** ____.

La mediana es ____ *porque está en el* ____.

Word bank: median mean mode order middle

Listen for: Students order to 4, 7, 7, 10, 12 and identify 7 as the median (middle value).

Level 2

Generalize: explain how to find the median when a data set has an EVEN number of values.

- With an even number of values, I ____ the two middle numbers.
- For example, with 4, 6, 8, 10 the median is ____ because ____.

Guided Examples

Example 1

Find the mean of: 10, 14, 8, 12, 16

Solution: $\text{Sum} = 10 + 14 + 8 + 12 + 16 = 60$. $\text{Mean} = 60 \div 5 = 12$.

Answer: A. 12

Example 2

Find the median of: 3, 7, 2, 9, 5

Solution: Ordered: 2, 3, 5, 7, 9. The middle value is 5.

Answer: A. 5

Example 3

Find the mode of: 4, 7, 4, 9, 2, 4, 8

Solution: 4 appears three times — more than any other value. $\text{Mode} = 4$.

Answer: A. 4

Write About the Math

The Writing Revolution

I can explain my reasoning using the words mean, median, mode, and outlier.

1. Kernel Sentence subject + verb

Model: Mean is the average. Add all the numbers, then divide by how many there are.

Media es el promedio. Suma todos los números y divide entre cuántos hay.

Write a kernel sentence about mean. Use a subject and a verb.

Escribe una oración base sobre media. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mean matters in math

Media importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mean matters in math because ____.

Media importa en matemáticas porque ____.

but
pero

Mean matters in math, but ____.

Media importa en matemáticas, pero ____.

so
entonces

Mean matters in math, so ____.

Media importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about mean.
Di un hecho verdadero sobre mean.

Mean ____.

Question
Pregunta

Ask a question about mean.
Haz una pregunta sobre mean.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about mean.
Muestra entusiasmo sobre mean.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with mean.
Dile a un compañero qué hacer con mean.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I would use the ____ **because it best shows the typical score.**

Usaría la ____ *porque muestra mejor el puntaje típico.*

The ____ **is the most common score, which is** ____.

La ____ *es el puntaje más común, que es* ____.

To find the median, I first ____ **the scores, then pick the** ____.

Para hallar la mediana, primero ____ *los puntajes y luego elijo el* ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. Find the mean of: 6, 8, 10, 12, 14

- A. 10
- B. 8
- C. 12
- D. 50

Show your work:

2. Data set: 4, 6, 6, 8, 10, 12. What is the median?

- A. 7
- B. 6
- C. 8
- D. 7.67

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A basketball player scored 15, 18, 16, 42, and 14 points in 5 games. Find the mean and median. Which better represents the player's typical game? Explain how the outlier affects each measure.

Sentence starter: Mean = _____. Median = _____. The _____ better represents a typical game because the outlier (_____) pulls the _____ toward it, making it _____ than most scores. The median is not affected because _____.

Show your work:

Reflect — Exit Ticket

Data set: 4, 7, 10, 7, 12. What is the median?

- A. 7
- B. 8
- C. 10
- D. 4

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 10 — *Sum = $6 + 8 + 10 + 12 + 14 = 50$. Mean = $50 \div 5 = 10$.*
2. **Try It 2:** A. 7 — *With 6 values, the median is the average of the 3rd and 4th values: $(6 + 8) \div 2 = 7$.*
3. **Exit Ticket:** A. 7 — *Ordered: 4, 7, 7, 10, 12. The middle (3rd) value is 7.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mean is the average. Add all the numbers, then divide by how many there are.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).